Cassava 3 — Data

Oct 26, 2026 · Anas Alsirafy

One dataset: the 2020 Kaggle Cassava Leaf Disease Classification set, 21,397 field photos in five classes. It is dominated by one class — cassava mosaic disease is 61.5% of images — so always guessing CMD already scores 61.5%. The duplicate screen found no duplicates.

Source

- Kaggle competition page; download with the Kaggle API after accepting the competition rules.
- Photos taken by farmers in Uganda on phones: cluttered backgrounds, mixed lighting, several leaves per frame. Treat it as field data, not lab data.
- The competition's own test set is hidden; all our numbers come from cross-validation on the 21,397 training images.

Classes

Counts match the class weights logged in training (weight = N / (5 × count)).

Index	Class	Images	Share	Class weight
0	Cassava Bacterial Blight (CBB)	1,087	5.1%	3.94
1	Cassava Brown Streak Disease (CBSD)	2,189	10.2%	1.95
2	Cassava Green Mottle (CGM)	2,386	11.2%	1.79
3	Cassava Mosaic Disease (CMD)	13,158	61.5%	0.33
4	Healthy	2,577	12.0%	1.66

The four classes other than CMD are the "minority classes" that the Exp 11 specialist targets.

Integrity screening

Every image was resized to 224×224 and hashed (MD5 for exact copies, 64-bit dHash for near copies, Hamming distance ≤ 5). Cassava had 0 exact and 0 near duplicates, so 21,397 images form 21,397 groups. Plain StratifiedKFold is therefore safe here, though the cross-dataset study used StratifiedGroupKFold for consistency.

Pre-processing and splits

- **CLAHE (from Exp 4 on):** LAB colour space, CLAHE on the L channel only (clipLimit 2.0, 8×8 tiles), back to RGB. Done once, offline, for all 21,397 images; the processed path is stored in a `clahe_path` column.
- **Resize:** 224×224 up to Exp 4, 384×384 from Exp 5 on.
- **Normalisation:** the backbone's own `preprocess_input` (e.g. EfficientNet), applied after augmentation.
- **Splits:** 5-fold StratifiedKFold, shuffled, fixed seed. Validation for callbacks is the held-out fold itself (see Known Issues).
- **Augmentation:** only on training batches, never on the validation fold.